INTRODUCTION TO PRINCIPLES OF ONCOLOGY

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S IV UNIT

- ONCOLOGY IS THE STUDY OF TUMOURS(BOTH BENIGN AND MALIGNANT), THEIR DEVELOPMENT, DIAGNOSIS, MANAGEMENT AND PREVENTION

- BENIGN TUMOURS GENERALLY DO NOT INVADE DEEPER TISSUES. THEY ARE ENCAPSULATED AND DO NOT METASTASISE
- MALIGNANT TUMOURS TEND TO INVADE DEEPER TISSUES, LACK A CAPSULE AND METASTASISE. CANCER CELLS LACK UNIFORMITY, DIFFER FROM THE PARENT CELLS IN ARRANGEMENT AND APPEARANCE, HAVE BIG AND PROMINENT NUCLEI AND NUCLEOLI. THE CHROMATIN LOOKS CONDENSED

TILL THE BASEMENT MEMBRANE IS PENETRATED, THE LESION IS CALLED 'CANCER IN SITU' OR A 'PRE MALIGNANT LESION'. AFTER THIS POINT, IT BECOMES AN INVASIVE CANCER

- MALIGNANT TUMOURS ARE SECOND ONLY TO HEART DISEASE WITH REGARD TO MORTALITY

TUMOUR NOMENCLATURE

- THE TERM 'CANCER' DENOTES MALIGNANT NEOPLASMS
- TUMOURS ARE NAMED AFTER THEIR TISSUE OF ORIGIN
- FOR BOTH BENIGN AND MALIGNANT TUMOURS,THE SUFFIX –*oma* IS ADDED TO THE NAME OF THE CELL OF ORIGIN. FOR MALIGNANT TUMOURS, IN ADDITION, THE EMBRYONAL TISSUE OF ORIGIN IS ALSO EXPRESSED
- TUMOURS OF EPITHELIAL ORIGIN ARE CALLED "CARCINOMAS" AND TUMOURS OF MESENCHYMAL ORIGIN ARE CALLED "SARCOMAS"
- EPITHELIAL TUMOURS MAY BE CALLED "ADENOCARCINOMA" OR "EPITHELIOMA" BASED ON THEIR HISTOLOGICAL APPEARANCE

TUMOUR NOMENCLATURE
PROGRESS AND RESPONSE TO THERAPY

- "COMPLETE RESPONSE" MEANS THERE IS NO DEMONSTRABLE TUMOUR
- "PARTIAL RESPONSE" MEANS A REDUCTION OF TUMOUR MASS BY GREATER THAN 50%
- "TUMOUR PROGRESSION" SIGNIFIES FAILURE OF THERAPY AND IS DEFINED AS A 50% OR GREATER INCREASE IN THE TUMOUR MASS

TUMOUR NOMENCLATURE PROGRESS AND RESPONSE TO THERAPY Contd.

- "LOCAL TREATMENT" MEANS MANAGEMENT OF THE PRIMARY TUMOUR
- "LOCO-REGIONAL TREATMENT" MEANS MANAGEMENT OF THE TUMOUR AND THE REGIONAL NODES
- "SYSTEMIC TREATMENT" MEANS MANAGEMENT OF THE METASTASIS

‘TUMOUR DOUBLING TIME’

- THE DOUBLING TIME OF A CANCER CELL IS ABOUT THREE DAYS.
- IT TAKES 90 DAYS AND 30 DOUBLINGS FOR A TUMOUR TO REACH A SIZE OF 1gm. AFTER THIS, IT TAKES ONLY 10 DOUBLINGS TO REACH A SIZE OF ABOUT 1 kg.!

CARCINOGENESIS

- *CARCINOGENESIS* IS THE PROCESS BY WHICH EXTERNAL FORCES CAUSE CANCER THROUGH GENETIC ALTERATION
  - THIS IS BY ***INITIATION*** AND ***PROMOTION***
  - ***INITIATION*** IS THE PROCESS BY WHICH A NORMAL CELL IS TRANSFORMED TO THE PREMALIGNANT STATE
  - ***PROMOTION*** IS THE PROCESS BY WHICH FURTHER CHANGES IN THE CELL RESULT IN THE FORMATION OF A TUMOUR

- AGENTS WHICH CAUSE CANCER ARE CALLED CARCINOGENS

- THEY MAY BE PHYSICAL, CHEMICAL OR INFECTIVE

SOME PHYSICO CHEMICAL AGENTS WHICH CAN CAUSE CANCER

POLYCYCLIC AROMATIC HYDROCARBONS

Eg. BENZOPYRENE AND 3-METHYLCHOLANTHRENE

- VINYL CHLORIDE

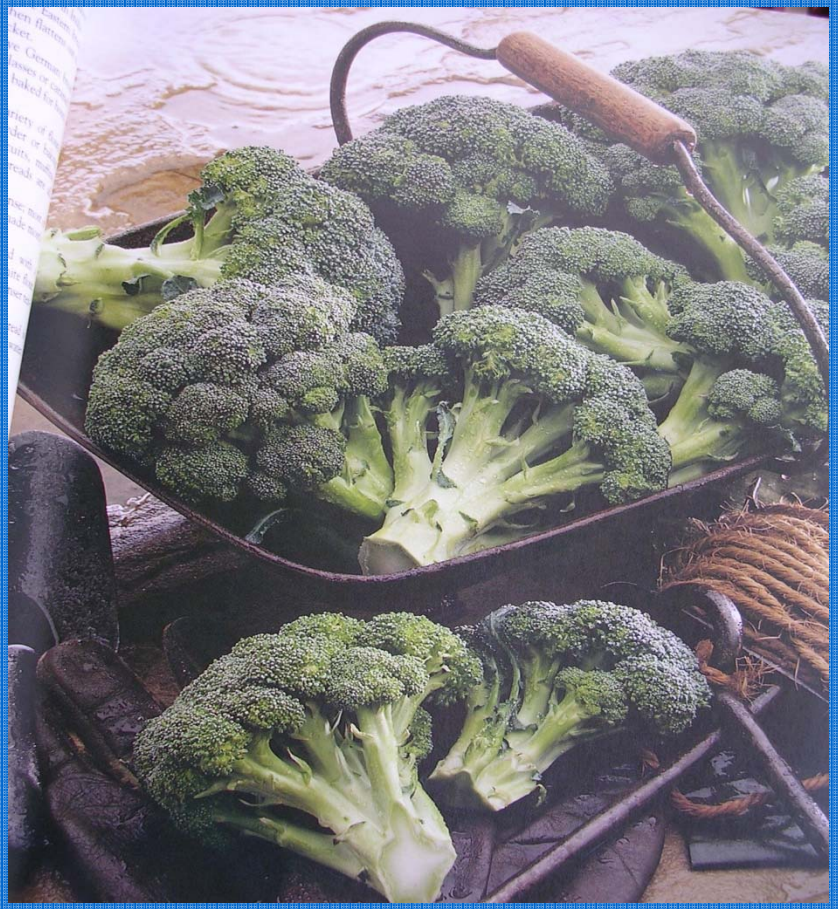
UV LIGHT, X-RAYS,RADIATION, AFLATOXIN,ASBESTOS, ALCOHOL, OESTROGEN

- INFECTIVE AGENTS ARE VIRUSES LIKE *SHOPE PAPILLOMA VIRUS* AND *PAPOVA VIRUS SV 40* IN MICE.
- IN HUMANS,VIRUSES LIKE HUMAN PAPILLOMA VIRUS, HEPATITIS-B VIRUS, AND EPSTEIN BARR VIRUS ARE IMPLICATED. CARCINOMA OF UTERINE CERVIX, NASOPHARYNGEAL CANCER, BURKITT'S LYMPHOMA, T-CELL LYMPHOMA AND KAPOSI'S SARCOMA MAY BE DUE TO VIRAL AETIOLOGY

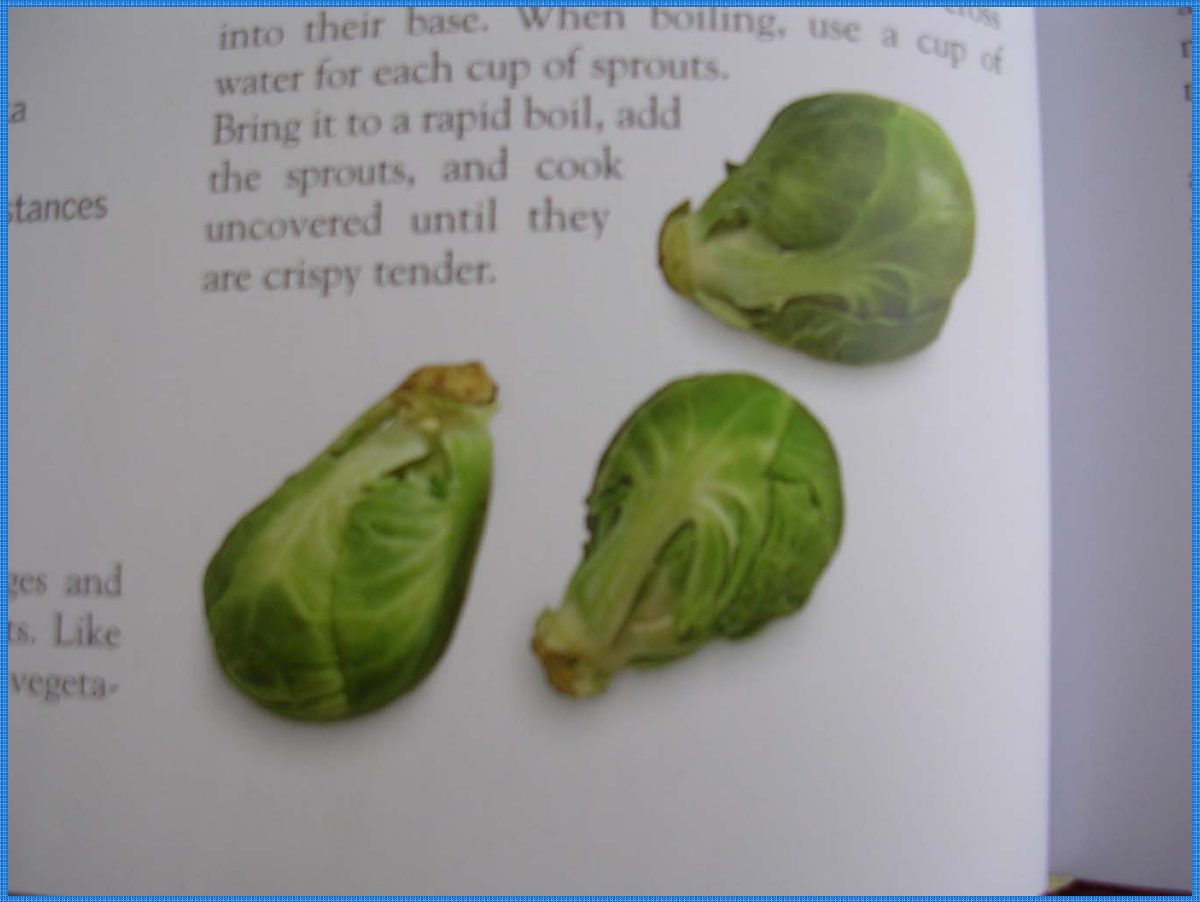
DIET

- HIGH CALORIE DIET CAN PREDISPOSE
- HEAVILY COOKED MEAT CONTAINS HETEROCYCLIC AMINES WHICH ARE CARCINOGENS
- SOME VEGETABLES LIKE BROCCOLI AND BRUSSELS SPROUT CONTAIN ANTIOXIDANTS AND 'DRUG METABOLISING ENZYMES' WHICH CAN METABOLISE HETEROCYCLIC AMINES

BROCCOLI



BRUSSELS SPROUT



MOLECULAR GENETICS OF CANCER

GENETIC MUTATION IS ESSENTIAL FOR FORMATION OF MALIGNANCY, BOTH SPORADIC AND INHERITED.

1) ONCOGENES

THEY ARE DERIVATIVES OF PROTO- ONCOGENES WHICH ARE NORMAL GENES ASSOCIATED WITH GROWTH AND FUNCTION. GENETIC ALTERATION OF ONCOGENES RESULTS IN THEIR ACTIVATION AND THIS IS CALLED 'GAIN OF FUNCTION'.

ALTERATION OF ONE COPY OR ALLELE OF THIS GENE IS ENOUGH TO BRING ABOUT THIS CHANGE AND THIS IS CALLED 'DOMINANT'.

EXAMPLES OF SOME ONCOGENES

- k-ras(PANCREATIC, COLORECTAL, ENDOMETRIAL)
- EGFR (GLIOMA, SQUAMOUS Ca.)
- c-myc (BURKITT'S LYMPHOMA, SMALL CELL Ca. OF LUNG)
- mdm-2(SARCOMAS)
- cycD1(BREAST)
- ERB-B2 (BREAST, STOMACH)

2) TUMOUR SUPPRESSOR GENES

TSG ARE RESPONSIBLE FOR PRODUCTION OF PROTEINS WHICH HAVE INHIBITORY FUNCTIONS ON CELL GROWTH, AND DIFFERENTIATION.

UNLIKE ONCOGENES, TSG HAVE TO BE INACTIVATED TO PRODUCE TUMOURIGENIC EFFECT AND THIS IS CALLED "LOSS OF FUNCTION".

BOTH ALLELES OF TSG HAVE TO BE INACTIVATED BY GENETIC ALTERATION TO PRODUCE AN ADVERSE EFFECT AND THIS IS CALLED "RECESSIVE".

EXAMPLES OF TUMOUR SUPPRESSOR GENES

- *APC (Adenomatous Polyposis Coli)* (COLORECTAL)
- *p53* (MORE THAN 50% OF ALL CANCERS)
- *RB1* (RETINOBLASTOMA, LUNG, PROSTATE, AND BREAST)
- *BRCA 1* (BREAST)
- *BRCA 2* (BREAST)
- *WT 1* (WILM'S TUMOUR)
- *SMAD 4* (PANCREATIC)
- *VHL* (KIDNEY, PHAEOCHROMOCYTOMA)

- SOME ONCOGENES ARE CAPABLE OF PRODUCING TUMOURS ON THEIR OWN, BUT GENERALLY, COLLABORATION BETWEEN MANY ONCOGENES AND TSG IS NECESSARY TO PRODUCE MALIGNANCY.

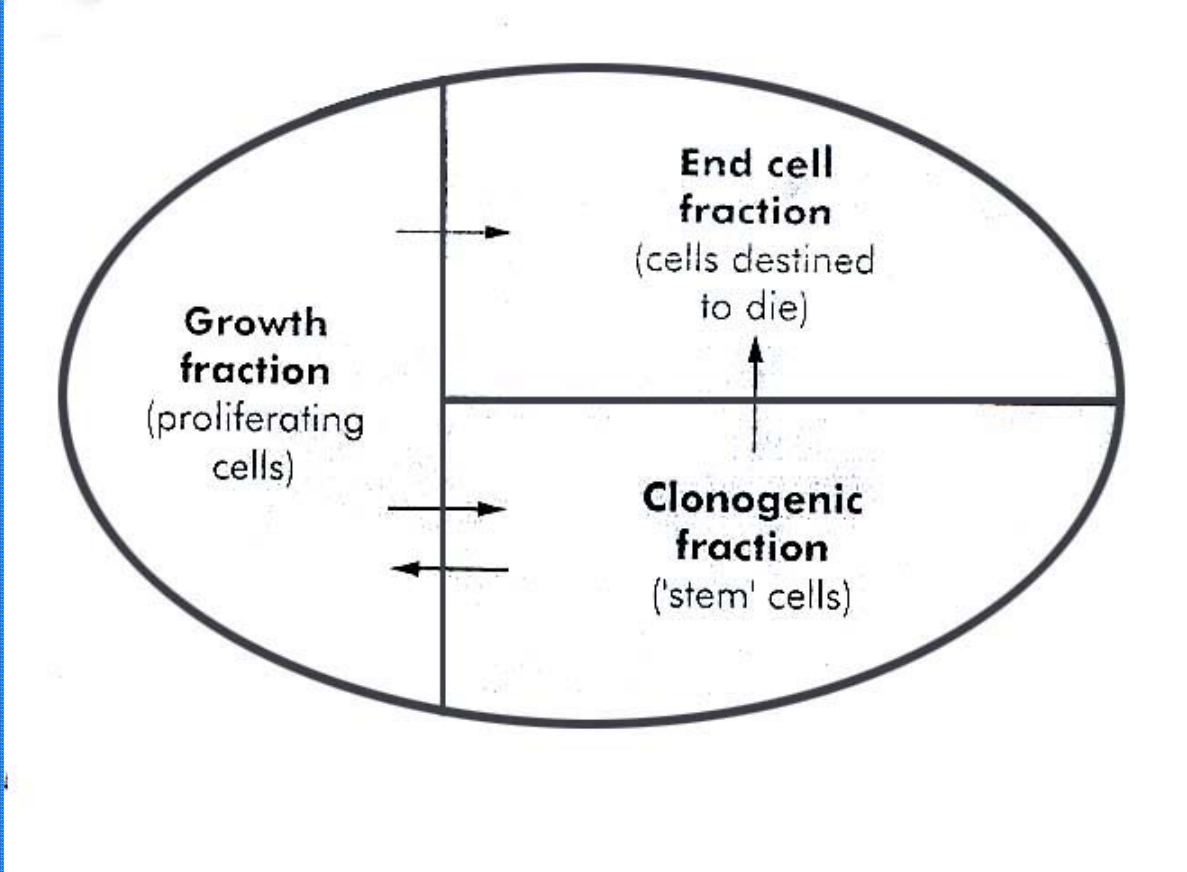
HEREDITARY CANCER SYNDROMES

- A FEW CANCERS LIKE RETINOBLASTOMA, AND COLORECTAL TUMOURS ARE TRANSMITTED TO THE OFFSPRING. THIS IS MOST OFTEN DUE TO MUTATION OF TSG
- DETECTION OF GERMLINE MUTATION IN THE OFFSPRING NEEDS SCREENING, SURVEILLANCE, TREATMENT AND GENETIC COUNSELLING
- OFFSPRING WHO HAVE NOT INHERITED THE GERMLINE MUTATION NEED ONLY REASSURANCE

THE KINETICS OF TUMOURS

- MALIGNANT TUMOURS CONSIST OF NOT ONLY CANCER CELLS, BUT ALSO OF FIBROUS TISSUE, FIBROBLASTS, MACROPHAGES, LEUCOCYTES AND BLOOD VESSELS. THESE ELEMENTS ARE ESSENTIAL FOR THE GROWTH AND SURVIVAL OF THE TUMOUR
- MALIGNANT CELLS WITHIN A TUMOUR CAN BE GROUPED INTO THREE FRACTIONS
    - 1) GROWTH FRACTION
    - 2) CLONOGENIC FRACTION
    - 3) END CELL FRACTION

TUMOUR KINETICS CONTINUED
CELL FRACTIONS



TUMOUR KINETICS CONTINUED

- CELLS OF *GROWTH FRACTION* ARE ACTIVELY DIVIDING
- CELLS OF *CLONOGENIC FRACTION* ARE NOT DIVIDING BUT HAVE THE CAPACITY TO DIVIDE IF STIMULATED
- CELLS OF *END CELL FRACTION* HAVE NO CAPACITY TO DIVIDE AND SO, THEY WILL DIE

This is important when considering chemotherapy. Chemotherapy will kill cells of growth fraction, but the clonogenic fraction can maintain the tumour by transferring cells into the growth fraction.

TUMOUR KINETICS CONTINUED
APOPTOSIS

- APOPTOSIS IS PROGRAMMED CELL DEATH.
  THIS IS THE COMMONEST FORM OF CELL DEATH IN TUMOURS

  APOPTOSIS CONSISTS OF A SERIES OF STEPS RESULTING IN CONDENSATION AND FRAGMENTATION OF CHROMATIN FOLLOWED BY SHRINKAGE OF THE CELL WHICH IS THEN TAKEN UP BY MACROPHAGES.

TUMOUR KINETICS CONTINUED

APOPTOSIS

RESISTANCE OF CELLS TO APOPTOSIS RESULTS IN GENETIC INSTABILITY AND MALIGNANCY.

APOPTOSIS IS THE FORM OF CELL DEATH FOLLOWING RADIOTHERAPY AND CHEMOTHERAPY. IF CANCER CELLS UNDER RADIO OR CHEMOTHERAPY DEVELOP RESISTANCE TO APOPTOSIS, THEY BECOME IMMUNE TO RADIO AND CHEMOTHERAPY.

TUMOUR KINETICS CONTINUED

- GROWTH CONTROL IN NORMAL TISSUES INVOLVES A BALANCE BETWEEN CELL DEATH AND CELL PROLIFERATION

- THUS, CELLS IN A MALIGNANT TUMOUR MAY BE DIVIDING SLOWER THAN SURROUNDING TISSUES, BUT THE RATE OF CELL DEATH MAY BE SLOW. THIS IS IMPORTANT WHEN CONSIDERING CHEMOTHERAPY

TUMOUR KINETICS CONTINUED
CELL CYCLE

CELL CYCLE IS THE GROWTH AND DIVISION OF THE CELLS.

THIS CONSISTS OF FIVE STAGES KNOWN AS

G 1, S, G 2, M AND G 0.

G 1 IS THE FIRST GAP PHASE DURING WHICH THE PROTEINS NECESSARY FOR NORMAL FUNCTIONING OF THE CELL ARE SYNTHESISED.

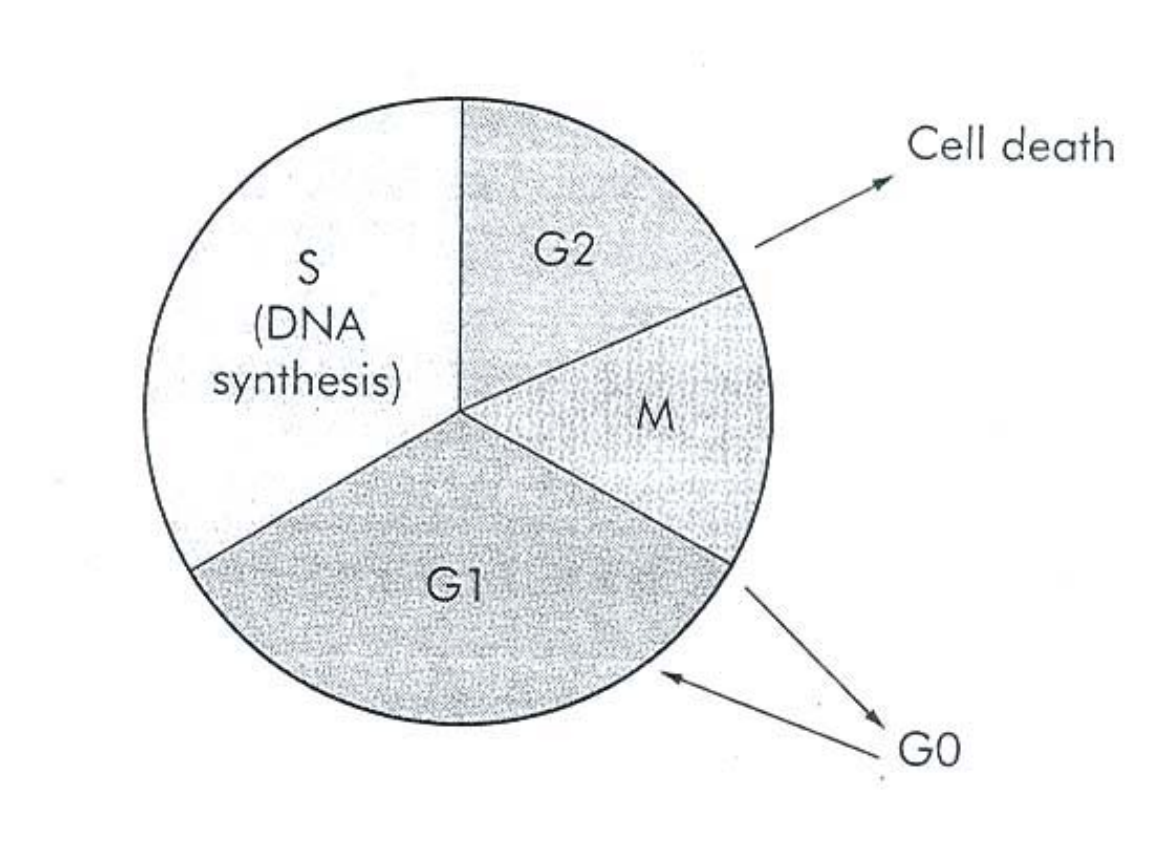
S PHASE IS THE SYNTHETIC PHASE DURING WHICH DNA REPLICATION OCCURS.

G 2 PHASE IS THE SECOND GAP PHASE.

M PHASE IS THE MITOSIS PHASE.

G 0 IS THE RESTING OR CLONOGENIC PHASE.

CELL CYCLE



TUMOUR KINETICS CONTINUED
CELL CYCLE

- A CELL CAN WITHDRAW FROM THE CELL CYCLE TO BECOME A RESTING OR CLONOGENIC CELL (G 0)

- IN A NORMAL CELL, G 0 CAN BE INDUCED BY WITHDRAWAL OF GROWTH FACTORS, WHEREAS THIS DOES NOT OCCUR IN MALIGNANT CELLS

TUMOUR KINETICS CONTINUED
CELL CYCLE

- NORMALLY, CELL CYCLE IS REGULATED IN SUCH A WAY THAT ONE PHASE OF THE CYCLE CANNOT START UNTIL THE PREVIOUS PHASE HAS BEEN COMPLETED CORRECTLY
- THE MECHANISMS WHICH GOVERN THIS PROPERTY ARE CALLED 'CHECK POINT CONTROLS'
- FOR EXAMPLE, IN A NORMAL CELL, IF DNA DAMAGE OCCURS, THE CELL WILL EITHER DIE VIA APOPTOSIS OR GO INTO CELL CYCLE ARREST(G 0) IN ORDER TO REPAIR THE DAMAGE
- THE PROTEIN p53 IS ESSENTIAL FOR THIS CHECK POINT CONTROL. DEFECTIVE p53 IS RESPONSIBLE IN MANY CANCERS BY ALLOWING THE PRODUCTION OF GENETICALLY UNSTABLE CELLS…

TUMOUR KINETICS

CELL CYCLE CONTINUED

- TWO MAJOR TYPES OF PROTEINS WHICH ARE INVOLVED IN CELL CYCLE
    - CYCLIN FAMILY
    - CYCLIN-DEPENDENT KINASES
- THESE TWO PROTEINS REGULATE THE DIFFERENT PHASES OF CELL CYCLE, AND THEIR ACTION CAN BE MODIFIED BY EXTRANEOUS FORCES LIKE GROWTH FACTORS, RESULTING IN MALIGNANCY

TUMOUR KINETICS CONTINUED

GOMPERTZIAN CURVE

SMALL TUMOURS GROW RAPIDLY WHEREAS LARGE TUMOURS GROW SLOWLY.

THIS MAY BE BECAUSE SMALL TUMOURS CONTAIN LARGER GROWTH FRACTIONS AND LARGE TUMOURS HAVE SMALLER GROWTH FRACTIONS

TUMOUR KINETICS CONTINUED

GOMPERTZIAN CURVE

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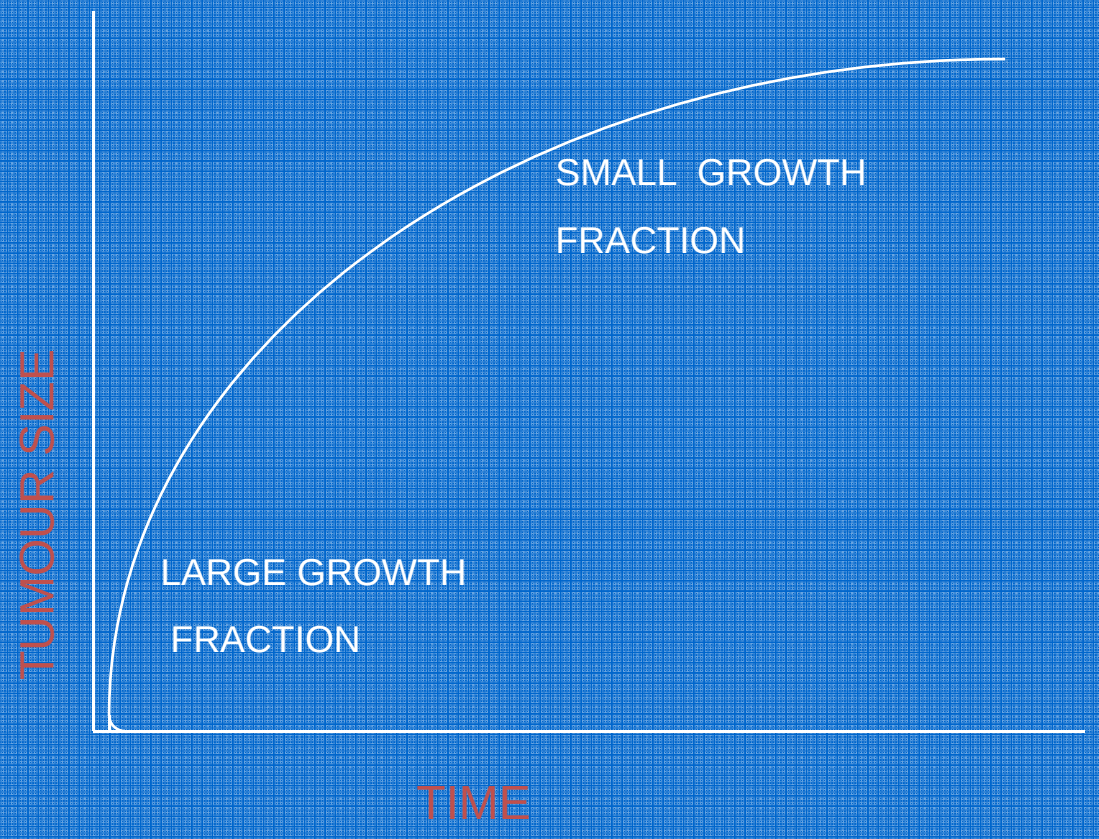


INVASION AND METASTASIS

INVASION IS PRODUCED BY INCREASED PRESSURE EXERTED BY RAPIDLY DIVIDING CELLS, LACK OF COHESION, ABNORMAL MOTILITY OF THE MALIGNANT CELLS OR BREAKDOWN OF THE EXTRACELLULAR MATRIX BY FACTORS PRODUCED BY THE TUMOUR.

RAPID PROLIFERATION IS BROUGHT ABOUT BY OVER EXPRESSION OF GROWTH FACTORS OR MUTATION OF GROWTH FACTOR RECEPTORS.

NORMAL CELL TO CELL ADHESION IS BROUGHT ABOUT BY SOME PROTEINS LIKE FIBRONECTIN AND LAMININ. THESE ARE KNOWN AS INTEGRINS.

LOSS OF ADHESION MAY LEAD TO MIGRATION OF THE MALIGNANT CELLS.

INVASION AND METASTASIS

CONTINUED

- INCREASED MOTILITY OF THE CANCER CELLS MAY BE DUE TO WEAKENING OF THE CYTOSKELETON BY SOME ONCOGENES OR BY ELABORATION OF PROTEOLYTIC ENZYMES LIKE CATHEPSINS, PLASMINOGEN ACTIVATORS AND METALLOPROTEINASES BY CANCER CELLS.
- METASTASIS OCCURS BY DISSEMINATION OF TUMOUR CELLS THROUGH BLOOD STREAM, LYMPHATICS OR COELOMIC SPACES.
- LIVER, LUNG AND BONE MARROW ARE MORE COMMONLY AFFECTED BY METASTASIS, WHEREAS SPLEEN, KIDNEY AND MUSCLE ARE NOT AFFECTED MUCH.

EFFECTS OF TUMOUR ON HOST

- LOCATION OF THE TUMOUR(Eg. PITUITARY ADENOMA)
- SECRETION OF HORMONES INDIGENOUS TO TISSUE OF ORIGIN
- SECRETION OF HORMONES NOT INDIGENOUS TO TISSUE OF ORIGIN

  (PARANEOPLASTIC SYNDROME. Eg. ACTH SECRETION BY SMALL CELL Ca. OF LUNG; INSULIN SECRETION BY HEPATOCELLULER Ca. LIVER

- CANCER CACHEXIA

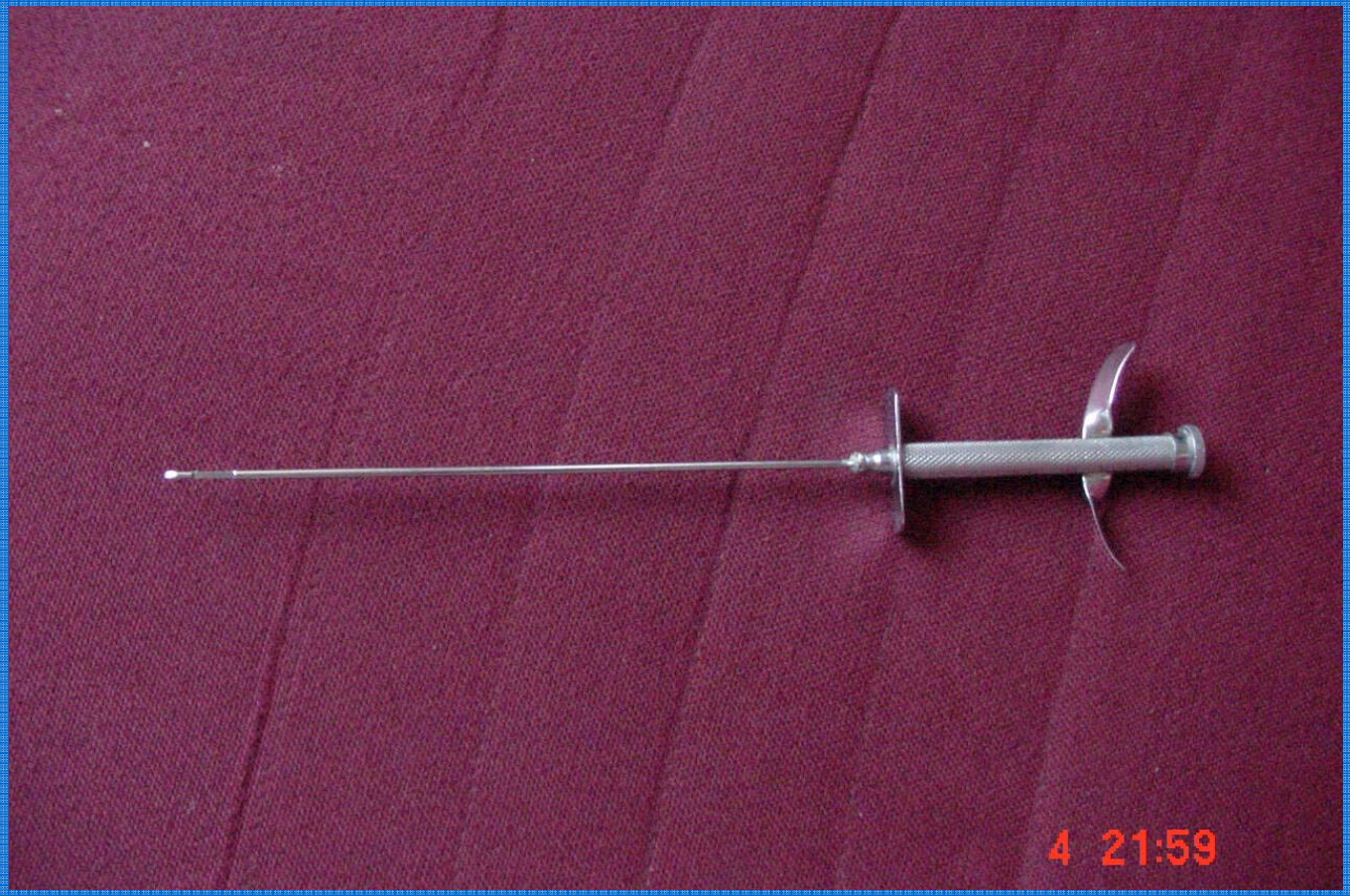
HOST RESPONSE TO TUMOUR

- CYTOTOXIC T-LYMPHOCYTES
- NATURAL KILLER CELLS
- ANTI TUMOUR ANTIBODIES
- MACROPHAGES

BIOPSY

- FNAC
- ENDOSCOPIC BIOPSY,BRUSH BIOPSY, PAPANICOLAU SMEAR, IMPRINT CYTOLOGY
- CORE BIOPSY (TRU-CUT NEEDLE BIOPSY)
- INCISION BIOPSY
- EXCISION BIOPSY
- FROZEN SECTION

TRU-CUT NEEDLE



TUMOUR GRADING AND STAGING

- TUMOUR GRADING AND STAGING ARE NECESSARY TO PREDICT THE PROGNOSIS, CHOOSE THERAPY AND COMPARE THE RESULTS FROM VARIOUS CENTERS.
- GRADING IS DONE DEPENDING ON THE NUMBER OF DIFFERENTIATED CELLS AND EXTENT OF MITOSIS.

THE GRADING IS DONE AS 1)WELL DIFFERENTIATED, 2)MODERATELY DIFFERENTIATED AND 3)POORLY DIFFERENTIATED (ANAPLASTIC). MOSTLY USEFUL FOR SARCOMAS

TUMOUR GRADING AND STAGING

CONTINUED

- STAGING IS DONE BY THE **UICC** SYSTEM (TNM SYSTEM) AND THE **AJCC** SYSTEM
- 'T' STANDS FOR TUMOUR SIZE, EXTENT AND INVASION. T_0 TO T_4 STAGES ARE THERE.
- 'N' STANDS FOR NODAL INVOLVEMENT. N_0 TO N_3 ARE THERE
- 'M' STANDS FOR METASTASIS. M_0 IS NO METASTASIS, M_1 IS PRESENCE OF METASTASIS.

LABORATORY DIAGNOSIS OF CANCER

- IMMUNO CYTOCHEMISTRY
  - USES MONOCLONAL ANTIBODIES TO DETECT THE CELL OF ORIGIN

    Eg. KERATIN INDICATES EPITHELIAL ORIGIN. PRESENCE OF DESMIN INDICATES MUSCLE ORIGIN

- FLOW CYTOMETRY
  - MEASURES CELL SURFACE ANTIGENS AND DNA CONTENT OF TUMOUR CELLS

LABORATORY DIAGNOSIS OF CANCER CONTINUED

- **TUMOUR MARKERS**
  - **SPECIFIC PROTEINS** PRODUCED BY CANCER Eg.PSA OF Ca.PROSTATE
  - **ONCOFETAL ANTIGENS** Eg. α- FETOPROTEIN IN Ca. OF LIVER AND NON SEMINOMATOUS TESTICULAR TUMOUR. CARCINOEMBRYONIC ANTIGEN IN Ca. OF PANCREAS, COLON, LUNG, STOMACH AND BREAST
  - **ISOENZYMES** Eg. ACID PHOSPHATASE IN Ca. PROSTATE
  - **HORMONES** Eg.HUMAN CHORIONIC GONADOTROPIN IN TROPHOBLASTIC TUMUORS AND CALCITONIN IN MEDULLARY CARCINOMA OF THYROID
  - **MUCINS AND GLYCOPROTEINS** Eg. CA 19-9 IN Ca. COLON AND PANCREAS, CA 15-3 IN Ca BREAST

SCREENING FOR CANCER

- SCREENING FOR A CANCER IS DONE IN AN AREA WHERE THAT PARTICULAR TYPE OF CANCER IS PREVALENT
- FOR SCREENING TO BE USEFUL, THE DISEASE MUST BE DIAGNOSABLE EARLY AND TREATMENT FOR EARLY DISEASE MUST BE AVAILABLE AND EFFECTIVE
- THE TEST MUST BE COST EFFECTIVE, SAFE AND ACCEPTABLE TO THE POPULATION. IT MUST BE SENSITIVE AND SPECIFIC